Caitao Zhan

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About Me

I do research in quantum information science with a focus on quantum networks.

Education

Aug. 2017 ~ Jan. 2024	Stony Brook University	Ph.D. Candidate in Computer Science Advisor: Himanshu Gupta
Aug. 2017 ~ Aug. 2022	Stony Brook University GPA: 3.92/4.0	M.S. in Computer Science
Sept. 2013 ~ Jun. 2017	China University of Geosciences GPA: 92/100	B.S. in Computer Science and Technology Rank: 1/122

Employment

Feb. 2024 ~ Present	Postdoctoral Scholar @ Argonne National Lab (Data Science and Learning Division) Quantum networks, Quantum network simulations (SeQUeNCe), Quantum Internet architecture
May 2021 ~ Aug. 2021	Software Engineering Intern @ Microsoft (Azure) `C#` development: Active Directory/Light-weight Directory Service (AD/LDS) Replay Tool Automation
Jun. 2018 ~ Jan. 2024	Research assistant @ Stony Brook University Wireless sensor networks, data center networks, quantum networks, and quantum sensor networks
Sep. 2017 ~ May 2018	Teaching assistant @ Stony Brook University Foundations of Computer Science, Analysis of Algorithms, Probability and Statistics for Data Science

Research Experience

May 2026 ~ Present	Quantum Networks: leverage AI to advance quantum networks
May 2025 ~ Present	Quantum Networks: Error Correction and DQC [20, 22]
May 2024 ~ Present	Quantum Networks: Hardware Modeling [16, 18]
Feb. 2024 ~ Present	Quantum Networks: Protocols and Architecture [13, 14, 17, 19, 21]
Jan. 2021 ~ Jan. 2024	Efficient Quantum Communication Networks. [8, 15] Design/implement routing algorithms/protocols for quantum networks. Bell states, GHZ, Graph states.
Sep. 2021 ~ Jan. 2024	Discrete Outcome Quantum Sensor Networks. [9, 12] Quantum state/channel discrimination, initial state optimization, semidefinite programming, theory.
Aug. 2022 ~ Sep. 2023	Quantum Sensor Network Algorithms for Transmitter Localization. [10] Quantum sensing, quantum state discrimination, quantum machine learning
Nov. 2019 ~ Mar. 2022	Intelligent Radio with Deep Learning. [5, 7, 11] Design/implement CNNs to solve wireless network problems: wireless localization & spectrum allocation. Reframe wireless problems to computer vision problems: image-to-image translation & object detection.
Mar. 2019 ~ Oct. 2019	Efficient Localization of Multiple Intruders in Shared Spectrum System. [3] Design/implement. Bayesian approach. Testbed(Odroid,Raspberry Pi,USRP,HackRF).
Dec 2018 ~ Sep. 2020	Datacenter Networks. [4] Multi-hop circuit switch scheduling. Greedy, approximation proof. Participate in implementation.
July 2018 ~ July 2019	Selection of Sensors for Efficient Transmitter Localization. [2, 6] Implement. Greedy, approximation proof. Bayesian approach. GPU acceleration.
Oct. 2015 ~ Sept. 2016	Optimization using Evolutionary Algorithms. [1] Design/implement. Ant colony optimization. Evolutionary algorithms for shape formation problems.

Selected Awards

China National Scholarship	2014, Chinese Ministry of Education, Top 0.2%
2nd Prize in University's ACM ICPC Cup (Programming contest)	2014, China University of Geosciences, Top 6%
Travel Grant (ACM IMC 2018)	2018, ACM Internet Measurement Conference
Best Poster Award (Graduate Research Day)	2022, Department of CS, Stony Brook University
Outstanding Poster Presentation Award (PRCS 2024)	2024, Argonne National Laboratory
Best Paper Award (QNET Track)	2025, IEEE Quantum Week
Postdoctoral Performance Award (PRCS 2025 co-chair)	2025, Argonne National Laboratory
LDRD Seed Award (PI)	2026, Argonne National Laboratory, $25K
R&D 100 Award (SeQUeNCe)	2026, R&D World

PhD Thesis

Transmitter Localization and Optimizing Initial State in Classical/Quantum Sensor Networks. Stony Brook University, 2024, PDF.

Paper to submit

[22] Sanjay Suresh, **Caitao Zhan**, Joaquin Chung, Rajkumar Kettimuthu, "SeQUeNCe.DQC: Full-stack Simulation of Distributed Quantum Computing"

[21] Joaquin Chung, Michal Hajdušek, ⋯, **Caitao Zhan**, Raj Kettimuthu, and Rodney Van Meter, "Cross-Validation of Open-Source Quantum Network Simulators"

Publication

[20] Sagar Patange, **Caitao Zhan**, Bikun Li, Joaquin Chung, Allen Zang, Liang Jiang, Rajkumar Kettimuthu, "Realistic Simulation of Quantum Repeater with Encoding and Classical Error Correction", IEEE QCE 2026, arXiv

[19] Joaquin Chung, Daniel Dilley, ⋯, **Caitao Zhan**, et al., "InterQnet: A Heterogeneous Full-Stack Approach to Co-designing Scalable Quantum Networks", IEEE Transactions on Quantum Engineering 2026, arXiv

[18] Hayden Miller, **Caitao Zhan**, Michael Bishof, Joaquin Chung, Han Xu, Prem Kumar, Rajkumar Kettimuthu, "Simulation of a Heterogeneous Quantum Network", QCNC 2026, arXiv

[17] Xinan Chen, **Caitao Zhan**, Joaquin Chung and Jeffrey Larson, "Routing Entanglement in Complex Quantum Networks Using GHZ States", QCNC 2026, arXiv

[16] Laura d'Avossa, **Caitao Zhan**, Joaquin Chung, Rajkumar Kettimuthu, Angela Sara Cacciapuoti, Marcello Caleffi, "Simulation of Quantum Transduction Strategies for Quantum Networks", IEEE International Conference on Quantum Computing and Engineering (QCE) 2025, PDF.

[15] X. Fan, **C. Zhan**, H. Gupta, C.R. Ramakrishnan, "Optimized Distribution of Entanglement Graph States in Quantum Networks", IEEE Transactions on Quantum Engineering (TQE) 2025. PDF.

[14] **Caitao Zhan**, Joaquin Chung, Rajkumar Kettimuthu, Allen Zang, Alexander Kolar, "Design and Simulation of the Adaptive Continuous Entanglement Generation Protocol", International Conference on Quantum Communications, Networking, and Computing (QCNC) 2025. PDF.

[13] Francesco Mazza, **Caitao Zhan**, Joaquin Chung, Rajkumar Kettimuthu, Marcello Caleffi, Angela Sara Cacciapuoti, "Simulation of Entanglement-Enabled Connectivity in QLANs using SeQUeNCe", IEEE International Conference on Communications (ICC) 2025. PDF.

[12] **C. Zhan**, H. Gupta, M. Hillery, "Optimizing Initial State of Detector Sensors in Quantum Sensor Networks", ACM Transactions on Quantum Computing (TQC). PDF.

[11] M. Ghaderibaneh, **C. Zhan**, H. Gupta, "DeepAlloc: CNN-Based Approach to Efficient Spectrum Allocation in Shared Spectrum Systems", IEEE Access 2024. PDF.

[10] **C. Zhan**, H. Gupta, "Quantum Sensor Network Algorithms for Transmitter Localization", IEEE International Conference on Quantum Computing and Engineering (QCE) 2023. PDF.

[9] M. Hillery, H. Gupta, **C. Zhan**, "Discrete Outcome Quantum Sensor Networks", Physical Review A (PRA). PDF.

[8] M. Ghaderibaneh, **C. Zhan**, C.R. Ramakrishnan, H. Gupta, "Efficient Quantum Network Communication using Optimized Entanglement-Swapping Trees", IEEE Transactions on Quantum Engineering (TQE) 2022. PDF.

[7] **C. Zhan**, M. Ghaderibaneh, P. Sahu, H. Gupta, "DeepMTL Pro: Deep Learning Based Multiple Transmitter Localization and Power Estimation", Elsevier Pervasive and Mobile Computing (PMC) 2022. PDF.

[6] A. Bhattacharya, **C. Zhan**, A. Maji, H. Gupta, S. Das, P. Djuric, "Selection of Sensors for Efficient Transmitter Localization", IEEE/ACM Transactions on Networking (TON) 2021. PDF.

[5] **C. Zhan**, M. Ghaderibaneh, P. Sahu, H. Gupta, "DeepMTL: Deep Learning Based Multiple Transmitter Localization", IEEE Symposium on a World of Wireless, Mobile and Multimedia Networks (WoWMoM) 2021. PDF.

[4] H. Gupta, M. Curran, **C. Zhan**, "Near-Optimal Multihop Scheduling in General Circuit-Switched Networks", ACM International Conference on emerging Networking EXperiments and Technologies (CoNEXT) 2020. PDF.

[3] **C. Zhan**, H. Gupta, A. Bhattacharya, M. Ghaderibaneh, "Efficient Localization of Multiple Intruders in Shared Spectrum System", ACM/IEEE Information Processing in Sensor Networks (IPSN) 2020. PDF.

[2] A. Bhattacharya, **C. Zhan**, H. Gupta, S. Das, P. Djuric, "Selection of Sensors for Efficient Transmitter Localization", IEEE International Conference on Computer Communications (INFOCOM) 2020. PDF.

[1] **C. Zhan** and C. Li, "Shape Formation in Games: a Probability-based Evolutionary Approach", 2016 International Conference on Computational Intelligence and Security. PDF.

Open-source Software

SeQUeNCe: developing a leading discrete event quantum network simulator. R&D 100 Award Winner.

Academic Services

`Co-chair` of:

- Argonne National Lab Postdoctoral Research and Career Symposium (PRCS '25)

`Program Committee` of the following conferences/workshops/tutorial:

- IEEE International Conference on Quantum Computing and Engineering (QCE '24, '25, '26)
- International Conference on Quantum Communications, Networking, and Computing (QCNC '25, '26, '27)
- International Conference on Computer Communications and Networks (ICCCN '26)
- Workshop on HPC/AI Integration with Quantum Computing/Networking (HAIQ '25, '26)
- Workshop on Quantum Network Simulations in conjunction with IEEE Quantum Week (QNSim '24, '25, '26)
- Tutorial on SeQUeNCe in conjunction with IEEE Quantum Week ('25, '26)

`Artifact Evaluation Committee` of the following conferences:

- ACM International Conference on Mobile Computing and Networking (MobiCom 2023)

`Reviewer` of the following journals:

- Elsevier The Journal of Networks and Computer Applications
- Elsevier Pervasive and Mobile Computing
- IEEE Internet of Things Journal
- IEEE Open Journal of the Communications Society
- IEEE Transactions on Networking
- IEEE Transactions on Communications
- IEEE Transactions on Quantum Engineering
- ACM Transactions on Quantum Computing
- Springer Nature npj Quantum Information

`Reviewer` of the following conferences:

- IEEE International Symposium on Information Theory (ISIT 2025)
- International European Conference on Parallel and Distributed Computing (EURO-PAR 2025)

Mentoring Experience

Grad students:

- Xiaojie Fan (Stony Brook University), 2023
- Laura Davossa (University of Naples Federico II), 2024
- Francesco Mazza (University of Naples Federico II), 2024

Undergrads:

- Sagar Patange (The University of Chicago), 2024-2026
- Hayden Miller (Brown University), 2025-2026
- Sanjay Suresh (University of Wisconsin-Madison), 2025-2026
- Joey Rinchiuso (The University of Chicago), 2026